NETWORKWALKS-TASHINGA-WK1-PM1-CYBERSECURITY-LAB-SETUP

An isolated VirtualBox lab with Kali Linux for ethical hacking and penetration-testing practice

Skill Cybersecurity | VirtualBox 7.2 | Kali Linux 2026.1 | Network 10.0.0.0/24 | Networkwalks Week 1

About

This is my Week 1 project for the **Networkwalks Cybersecurity** programme (task **WK1-PM1**). I built a self-contained testing lab on my own laptop: VirtualBox as the hypervisor, Kali Linux as the attacking machine, and a private NAT Network that every future lab machine can join.

Everything here is documented step by step with the screenshots I took while building it, including the mistakes I hit and how I fixed them.

> **Ethical use only.** Everything in this lab is confined to machines I own inside the lab. None of it may be used against systems without explicit written permission.

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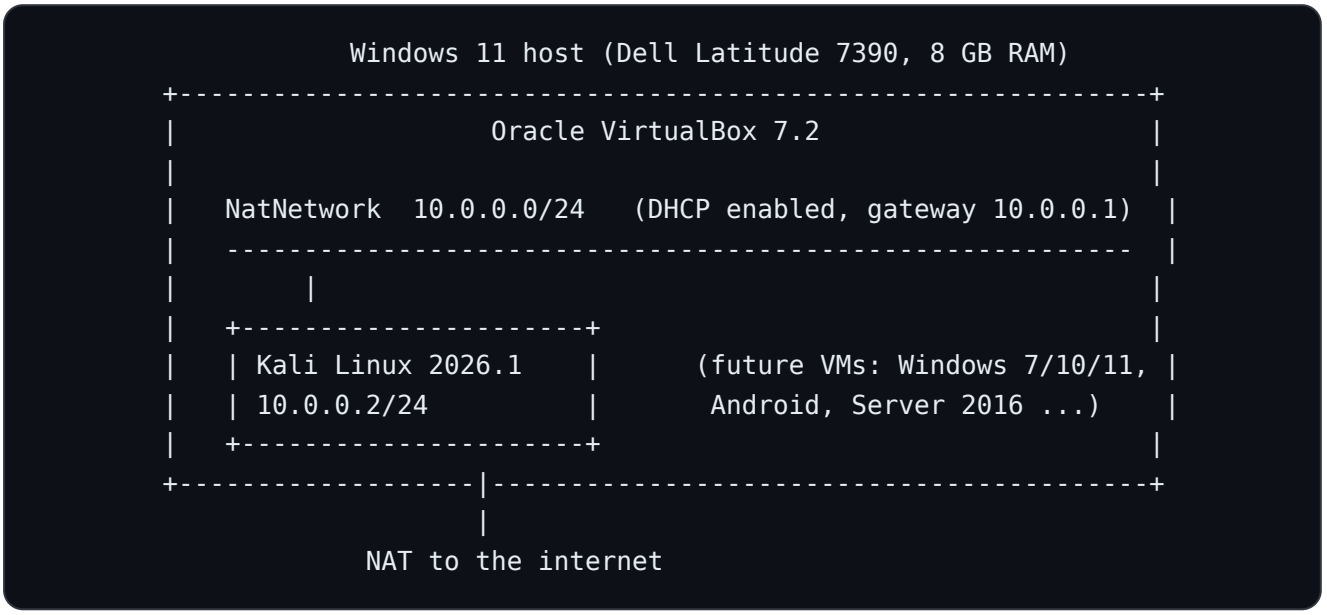
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Task requirements

Requirement	Status
VirtualBox as the base (latest recommended version)	Done
Kali Linux as the attacking machine	Done
NAT Network in subnet 10.0.0.0/24	Done
Clipboard and file drag-and-drop enabled	Done (bidirectional)
Host `Downloads` folder shared with Kali	Done (auto-mount, full access)
Kali IP address is 10.0.0.2/24	Done
Kali has full internet access	Done (0% packet loss)
Snapshot of the working VM	Done

Lab design



Why a **NAT Network** rather than plain NAT: machines on the same NAT Network can talk to each other and still reach the internet, which is exactly what a multi-VM practice lab needs. Plain NAT gives every VM its own isolated network.

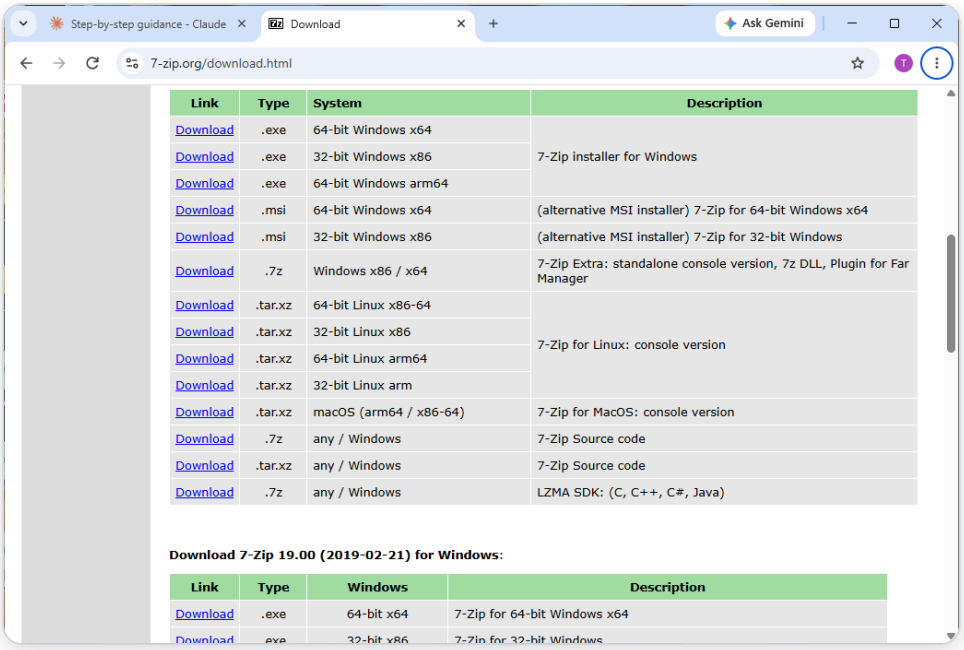
Lab configuration

Component	Configuration
Host OS	Windows 11
Host hardware	Dell Latitude 7390, 8 GB RAM
Hypervisor	Oracle VirtualBox 7.2
Attacking OS	Kali Linux 2026.1 (VirtualBox amd64 image)
Kali memory	2048 MB
Virtual network	NAT Network named `NatNetwork`
Network address	10.0.0.0/24 (DHCP enabled)
Kali IP address	10.0.0.2/24 (static)
Default gateway	10.0.0.1
DNS server	8.8.8.8
Adapter type	Intel PRO/1000 MT Desktop (82540EM)
Promiscuous mode	Allow All
Clipboard / drag-and-drop	Bidirectional
Shared folder	Host `Downloads`, auto-mount, full access

Setup walkthrough

Step 1. Install 7-Zip

Kali is distributed as a compressed archive, so 7-Zip is needed to extract the VM image. Download: https://7-zip.org/download.html. I used the 64-bit Windows x64 installer.



Link	Type	System	Description
Download	.exe	64-bit Windows x64	7-Zip installer for Windows
Download	.exe	32-bit Windows x86	
Download	.exe	64-bit Windows arm64	
Download	.msi	64-bit Windows x64	(alternative MSI installer) 7-Zip for 64-bit Windows x64
Download	.msi	32-bit Windows x86	(alternative MSI installer) 7-Zip for 32-bit Windows
Download	.7z	Windows x86 / x64	7-Zip Extra: standalone console version, 7z DLL, Plugin for Far Manager
Download	.tar.xz	64-bit Linux x86-64	7-Zip for Linux: console version
Download	.tar.xz	32-bit Linux x86	
Download	.tar.xz	64-bit Linux arm64	
Download	.tar.xz	32-bit Linux arm	
Download	.tar.xz	macOS (arm64 / x86-64)	7-Zip for MacOS: console version
Download	.7z	any / Windows	7-Zip Source code
Download	.tar.xz	any / Windows	7-Zip Source code
Download	.7z	any / Windows	LZMA SDK: (C, C++, C#, Java)

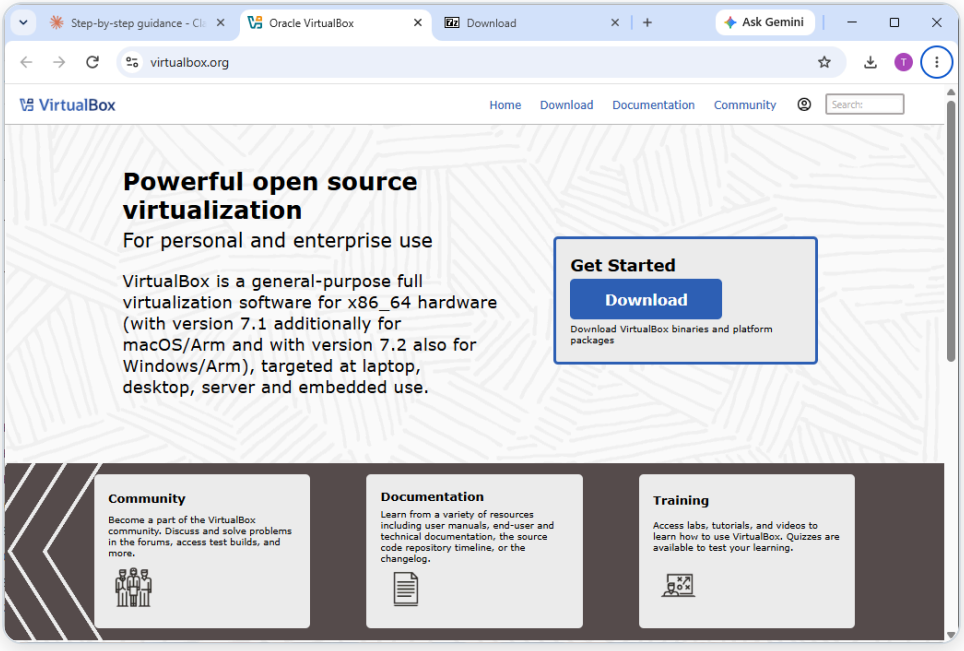
Download 7-Zip 19.00 (2019-02-21) for Windows:

Link	Type	Windows	Description
Download	.exe	64-bit x64	7-Zip for 64-bit Windows x64
Download	.exe	32-bit x86	7-Zip for 32-bit Windows

7-Zip download page

Step 2. Install VirtualBox

VirtualBox is the hypervisor that runs the lab. Download: https://virtualbox.org/wiki/Downloads. Choose the Windows hosts installer and accept the defaults.



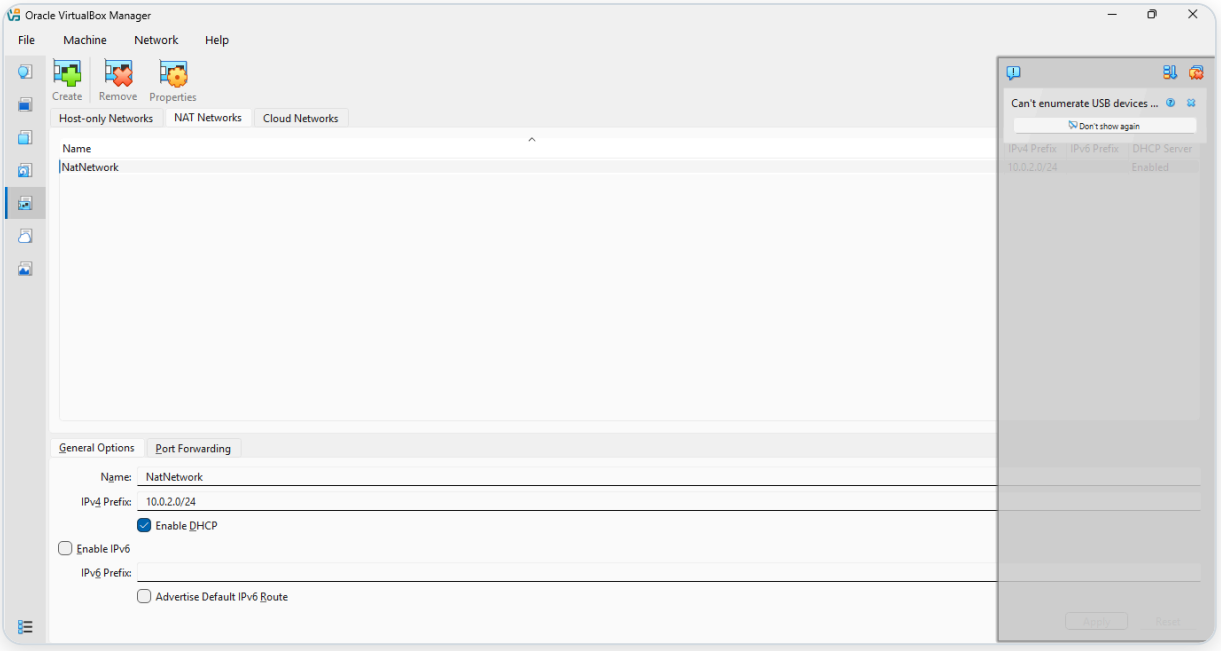
VirtualBox download page

Step 3. Create the NAT Network (10.0.0.0/24)

1. In VirtualBox Manager, open **File** → **Tools** → **Network**.
2. Open the **NAT Networks** tab and click **Create**.

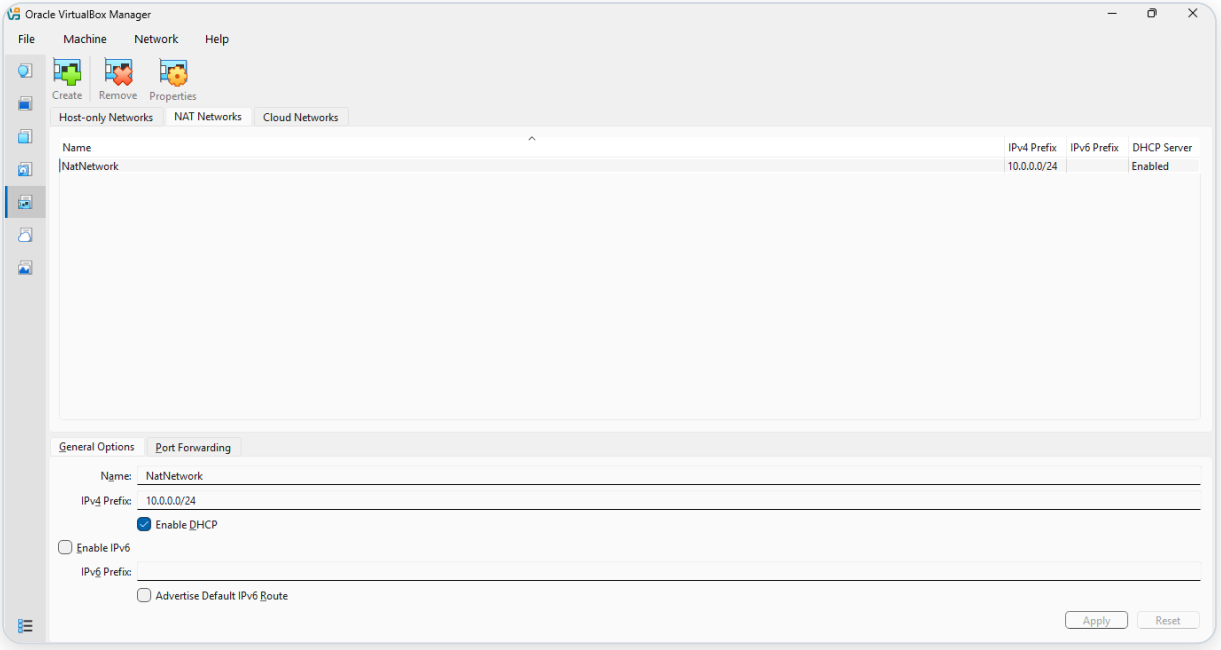
3. Set **Name** to `NatNetwork`, **IPv4 Prefix** to `10.0.0.0/24`, and keep **Enable DHCP** ticked.
4. Click **Apply**.

VirtualBox creates the network with its default prefix of `10.0.2.0/24`, so the prefix must be edited:



NAT Network created with the default 10.0.2.0/24 prefix

After the change the list shows `10.0.0.0/24` with the DHCP server enabled:

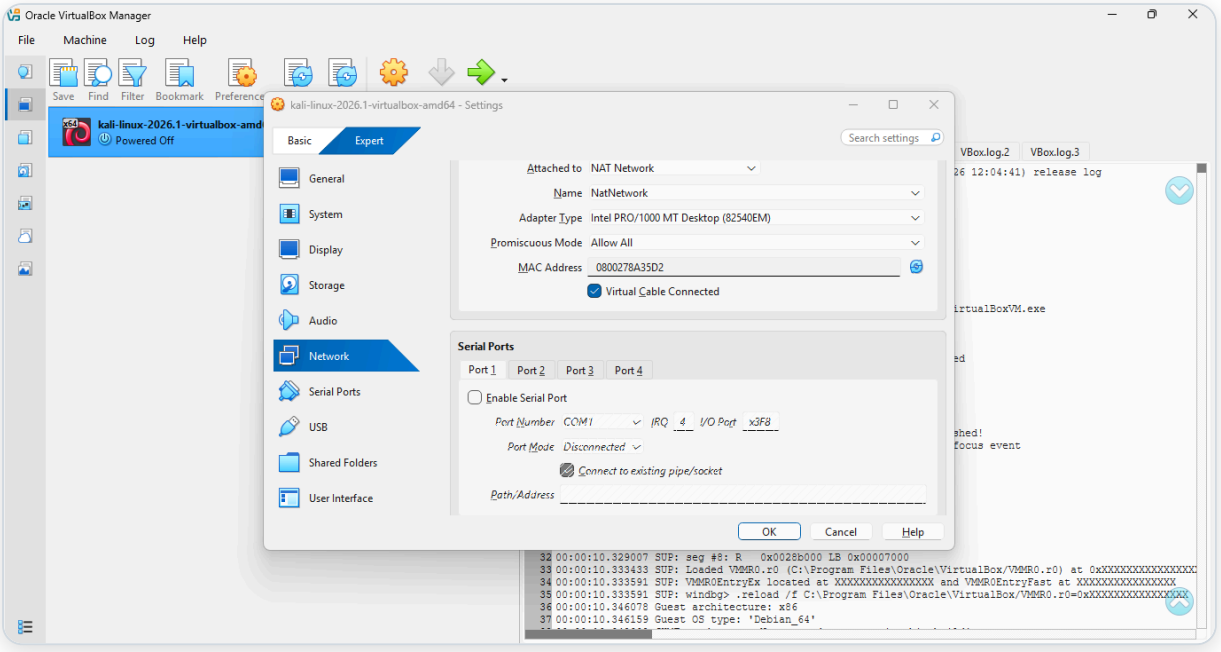


NAT Network with prefix 10.0.0.0/24

Step 4. Import Kali Linux and attach it to the NAT Network

Download the Kali VirtualBox image from https://kali.org/get-kali, extract it with 7-Zip, and add it to VirtualBox. Then open the VM's **Settings** → **Network** → **Adapter 1**:

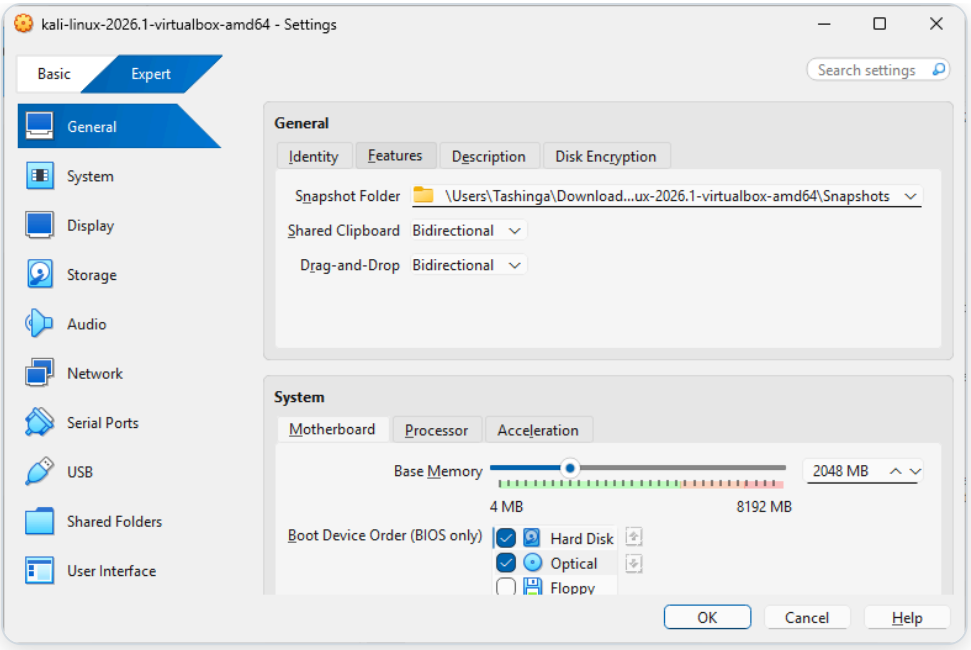
Setting	Value
Attached to	NAT Network
Name	NatNetwork
Promiscuous Mode	Allow All



Kali VM network adapter settings

Step 5. Enable the shared clipboard and drag-and-drop

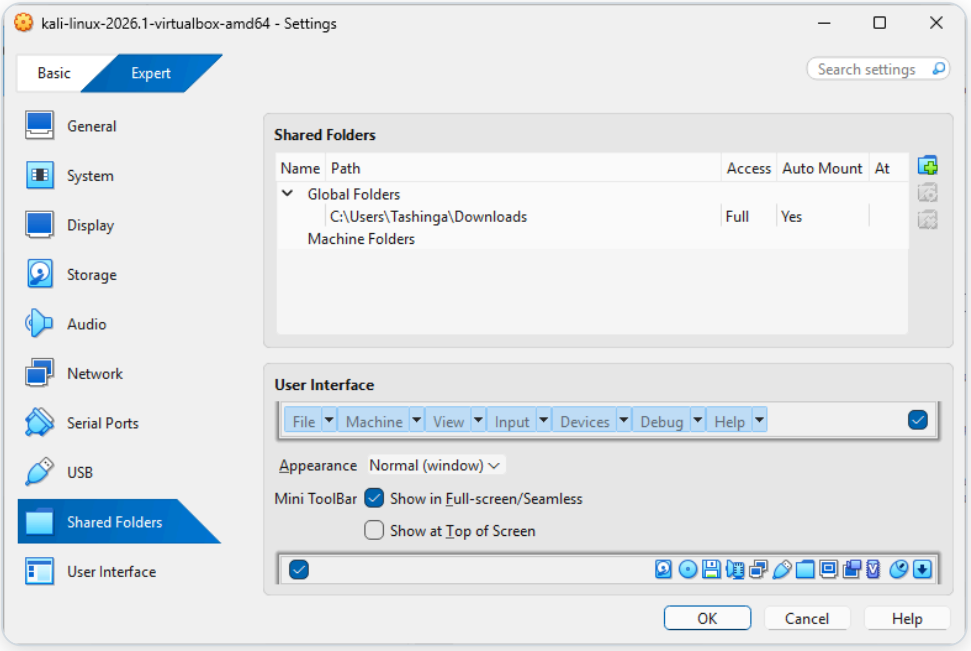
In **Settings** → **General** → **Features**, set **Shared Clipboard** and **Drag-and-Drop** to **Bidirectional**. In VirtualBox 7.2 these options sit under the Features tab.



Clipboard and drag-and-drop settings

Step 6. Share the host Downloads folder

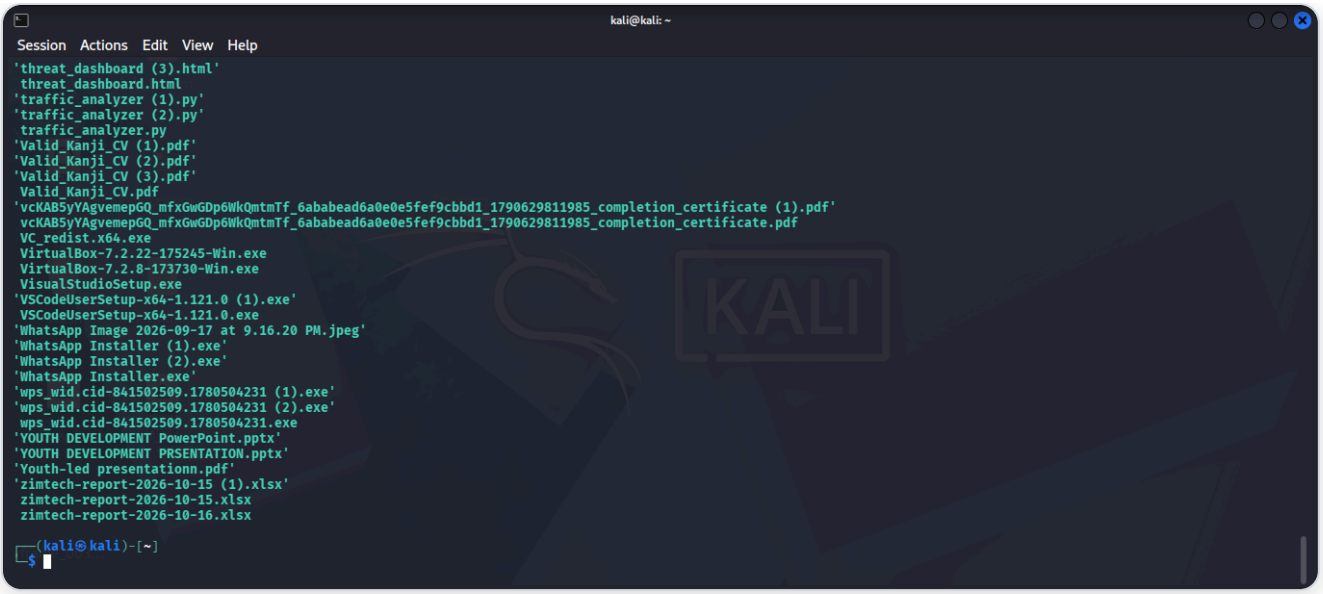
In **Settings** → **Shared Folders**, add the Windows `Downloads` folder with **Auto-mount** ticked and **Read-only** unticked (full access).



Shared Folders settings

Inside Kali the share appears as `sf_Downloads` and is mounted under `/media/`:

```
ls /media/sf_Downloads
```



Windows Downloads folder listed from inside Kali

Step 7. Set the static IP address in Kali

Run these in a Kali terminal (also available as `scripts/set-static-ip.sh`):

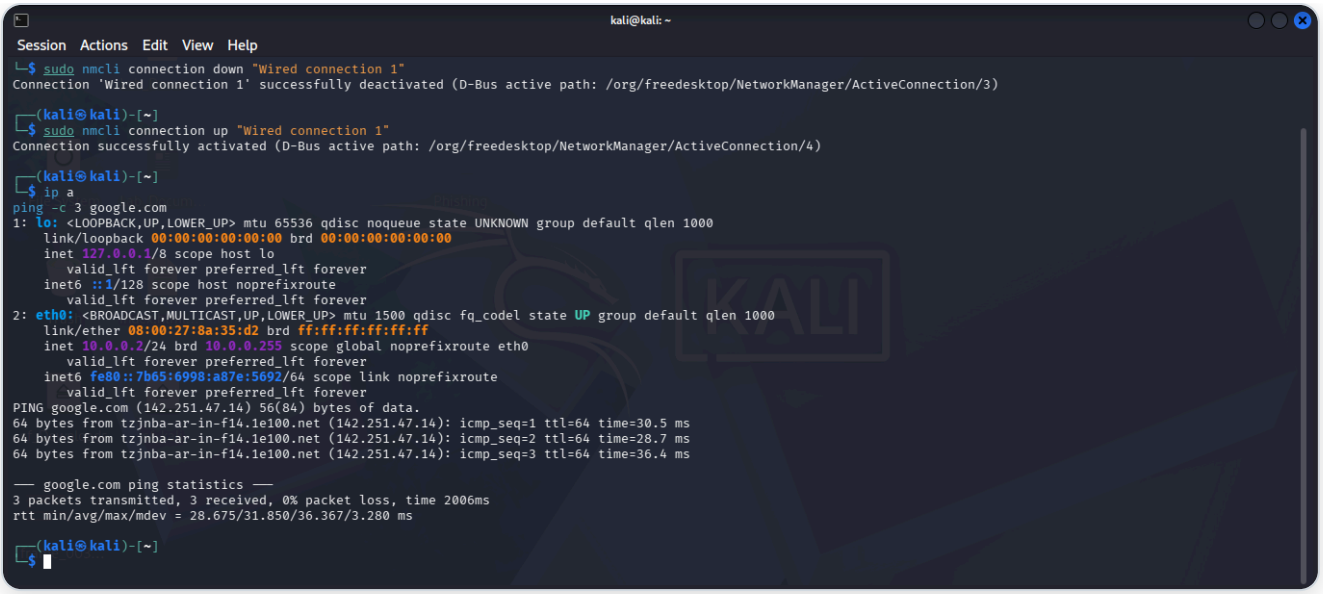
```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.0.0.2/24 \
  ipv4.gateway 10.0.0.1 \
  ipv4.dns 8.8.8.8 \
  ipv4.dad-timeout 0

sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

`ipv4.dad-timeout 0` turns off the duplicate-address-detection delay, a known fix for connectivity problems on VirtualBox 7 with recent Kali releases. If the DNS server does not respond, try the gateway (`10.0.0.1`) as the DNS server instead.

Step 8. Verify the IP address and internet access

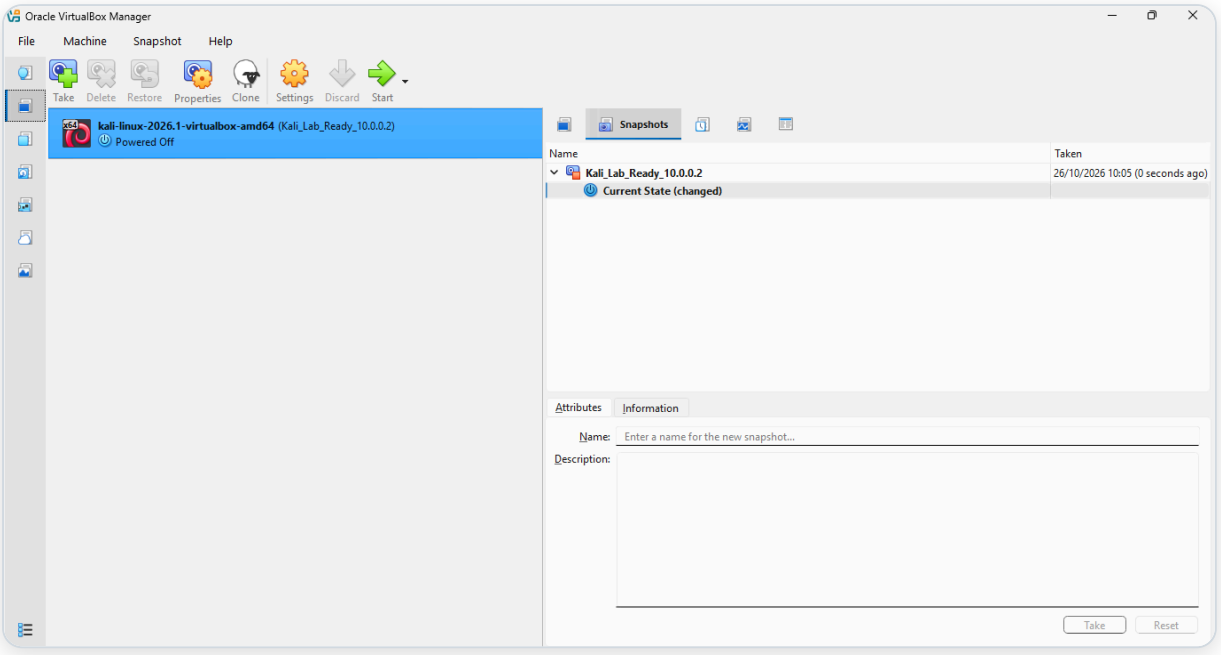
```
ip a
ping -c 3 google.com
```



ip a and ping results

Step 9. Take a snapshot

With Kali powered off, open the VM's **Snapshots** view and take a snapshot named `Kali_Lab_Ready_10.0.0.2`. It is the known-good starting point I can restore at any time.



Snapshot taken

Verification results

Check	Result
`eth0` address	`10.0.0.2/24`, broadcast `10.0.0.255`
Ping to google.com	3 sent, 3 received, **0% packet loss**
Average round-trip time	about 32 ms
Shared folder	Windows Downloads files visible in `/media/sf_Downloads`
Snapshot	`Kali_Lab_Ready_10.0.0.2` present

Problems I ran into

Problem	Cause	Fix
Kali had `10.0.2.15` instead of `10.0.0.2`	VM adapter was on plain NAT, so VirtualBox's default network was used	Attached the adapter to **NAT Network** and set the static IP
`Name` dropdown showed "Not selected" and OK was greyed out	The NAT Network had not been created yet	Created it under **File → Tools → Network → NAT Networks**
NAT Network came up as `10.0.2.0/24`	VirtualBox default prefix	Edited the **IPv4 Prefix** to `10.0.0.0/24` and applied
Clipboard and drag-and-drop options hard to find	In VirtualBox 7.2 they moved to **General → Features**	Set both to Bidirectional there
Add Share dialog had no "Access" dropdown	Newer versions use a **Read-only** checkbox	Left Read-only unticked for full access

Repository structure

```
NETWORKWALKS-TASHINGA-WK1-PM1-CYBERSECURITY-LAB-SETUP/
├── README.md                          This document
├── LICENSE                            MIT licence
├── docs/
│   └── WK1-PM1_Lab_Setup_Report.docx  Full submission report (Word)
├── screenshots/                       Evidence for every step
└── scripts/
    └── set-static-ip.sh               Sets the Kali static IP with nmcli
```

Next steps

- Add Windows 11, Windows 10, Windows 7 and Android VMs on the same NAT Network (planned addresses: 10.0.0.11, 10.0.0.10, 10.0.0.7 and 10.0.0.9).
- Ping-test every machine from every other machine and snapshot each one.
- Use the lab for reconnaissance, scanning and vulnerability-assessment exercises, always against my own lab machines only.

Tools and resources

- 7-Zip: https://7-zip.org/download.html
- VirtualBox: https://virtualbox.org/wiki/Downloads
- Kali Linux: https://kali.org/get-kali

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Programme	Cybersecurity at Networkwalks
Week	01
Project	WK1-PM1, Cybersecurity and Pentesting Lab Setup
Repository	GitHub

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